

Ting-Hsuan Chen

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Research Interests

Generative AI for embodied intelligence: video diffusion and world models for scaling robot data, physically consistent generation, world-action models that turn prediction into action, and humanoid and multi-agent embodied systems that generalize to unseen situations and coordinate with other agents.

Education

University of Southern California

Aug 2024 – present

PhD in Computer Science; MS in Computer Science | GPA: **3.92/4.0**

- Advised by Prof. Yue Wang; supported by the **Viterbi School Taiwan Global Fellowship**

National Chung Hsing University

Sept 2019 – June 2023

Department of Applied Mathematics (DAM) - Computer Science Group

- Final Average **96.6%**, GPA **4.27/4.3**; ranked 1st in all eight semesters; Valedictorian
- Admitted as the representative of the College of Science to the Phi Tau Phi Honor Society

Publications (*: Equal contribution, †: Equal advising)

PwP: Panoramas without Panoramas

Under review

Camera-Controlled 360° Video from Any View via Perspective-Only Adaptation

Ting-Hsuan Chen^{*}, Louison Lu^{*}, Qi Pan^{*}, Hengyuan Zhang, Xinyu Huang, Cho-Ying Wu, Yue Wang, Liu Ren
[Project page](#) [🔗](#)

CoLM: Controllable Text-Driven Humanoid Loco-Manipulation Generation

Under review

Han Jiang^{*}, Siheng Zhao^{*}, Ting-Hsuan Chen, Yue Wang
[Project page](#) [🔗](#)

TimePre: Bridging Accuracy, Efficiency, and Stability in Probabilistic Time-Series Forecasting

TMLR 2026

Lingyu Jiang^{*}, Lingyu Xu^{*}, Peiran Li, Dengzhe Hou, Qianwen Ge, Dingyi Zhuang, Shuo Xing, Wenjing Chen, Xiangbo Gao, Ting-Hsuan Chen, Xueying Zhan, Xin Zhang, Ziming Zhang, Zhengzhong Tu, Michael Zielewski, Kazunori Yamada, Fangzhou Lin
[arXiv](#) [🔗](#)

Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion

CVPR 2026

Ting-Hsuan Chen^{*}, Ying-Huan Chen^{*}, Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo[†], Yu-Lun Liu[†], Yue Wang[†], Liu Ren
[Project page](#) [🔗](#) [arXiv](#) [🔗](#)

Dino-Diffusion Modular Designs Bridge the Cross-Domain Gap in Autonomous Parking

Under review

Zixuan Wu, Hengyuan Zhang, Ting-Hsuan Chen, Yuliang Guo, David Paz, Xinyu Huang, Liu Ren
[arXiv](#) [🔗](#) [Code](#) [🔗](#)

MoonSim: A Photorealistic Lunar Environment Simulator

Under review

Ting-Hsuan Chen^{*}, Henghui Bao^{*}, Ziyu Chen^{*}, Haozhe Lou, Ge Yang, Zhiwen Fan, Marco Pavone, Yue Wang
[Project page](#) [🔗](#)

NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing

NeurIPS 2024

Ting-Hsuan Chen, Jiewen Chan, Hau-Shiang Shiu, Shih Han Yen, Changhan Yeh, Yu-Lun Liu

[Project page](#) [arXiv](#) [Code](#) [Demo](#)

DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models

arXiv 2024

Changhan Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Ting-Hsuan Chen, Yu-Lun Liu

[Project page](#) [arXiv](#) [Code](#) [Demo](#)

Experience

Scene Understanding/GenAI – Research Intern

Sunnyvale, United States

Bosch Center for Artificial Intelligence

May 2025 – present

- Co-first author of **Pantheon360** (CVPR 2026): a 3D-aware 360° video diffusion framework that leverages an explicit point cloud cache to ensure geometric consistency and precise trajectory control for digital twin generation.
- Co-first author of **PwP** (under review): adapts panoramic video generators to new domains using only perspective multi-view videos, generating camera-controlled 360° video from any initial view.
- Engineered a diffusion-based planning system with DINOv2 perception (**Dino-Diffusion**), enabling robust zero-shot generalization in challenging environments.
- Developing controllable driving behavior simulation that follows goal and sketch prompts without sacrificing safety, and object-centric video generation with accurate physics, fine-grained text control over individual objects, and strong generalization.

PhD Student/Research Assistant

Los Angeles, United States

University of Southern California

Aug 2024 – present

Advised by Prof. Yue Wang. My research turns generative priors into physical intelligence, along three thrusts:

- Data scaling**: using video diffusion and world models to generate robot data beyond teleoperation.
- Physical consistency**: pushing generated content to obey real-world physics, not merely look plausible.
- World-action coupling**: how world models turn prediction into action, via inverse dynamics, joint video-action denoising, or action-only inference with video co-training.

MS Student (Spring 2024)/Research Assistant

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

Jan 2024 – June 2024

- Entered Prof. Yu-Lun Liu's laboratory and led a Google-sponsored industry-academia collaboration project.
- Published **NaRCan** as first author at NeurIPS 2024 within five months.
- Collaborated with MediaTek to develop the first zero-shot video restoration framework utilizing diffusion-based image restoration models (**DiffIR2VR-Zero**).

R&D Engineer

Taipei, Taiwan

Foxconn, Hon Hai Precision Industry

July 2023 – Dec 2023

- Developed the company's first patented ECG waveform recognition system within three months, integrating AI, computer vision, and signal processing technologies.
- Built an AI-based data cleansing and classification system for medical data, together with its Django APIs.
- Served as a technical mentor for interns and represented the company at medical conferences.

Patents

System and Method of Modular Autonomous Parking Pipeline Using Visual Foundation Models and Diffusion Planners

U.S. Patent Application

Hengyuan Zhang, Zixuan Wu, Ting-Hsuan Chen, Yuliang Guo, David Paz, Xinyu Huang, Liu Ren

Assigned to Robert Bosch GmbH. A modular autonomous parking pipeline that combines visual foundation models with diffusion-based planners (the technology behind Dino-Diffusion).

Data Analysis Method, Apparatus, Electronic Device and Storage Medium

TW113141155, 2024

Ting-Hsuan Chen

A data analysis method based on feature waveform analysis that improves recognition accuracy through convolution and segmentation operations.

Projects

DreamMesh

[GitHub](#), 2025

Ting-Hsuan Chen, Cameron Smith, Jiageng Mao, Daniel Wang. A Blender plugin that transforms text or image inputs into complete 3D scenes with rigged objects, AI-generated backgrounds, and automatic object placement.

GPU-SplineTransformer

[GitHub](#), 2022

A GPU-based SplineTransformer that accelerates large-scale B-spline basis conversion by over $25\times$ compared with CPU-based solutions.

Open-Source Contributions

ViPE (NVIDIA Toronto AI Lab)

Aug 2025

Fixed multi-view SLAM initialization; the patch was merged into the official repository ([Issue #11](#), [PR #16](#)).

Academic Service

Conference Reviewer: NeurIPS, ICLR, CVPR

2025 – present

Honors & Awards

CVPR 2026 Broadening Participation Scholarship

Apr 2026

Viterbi School Taiwan Global Fellowship, USC

Mar 2026

Viterbi Conference & Research Fund Award, USC

Dec 2024

NeurIPS 2024 Scholar Award

Oct 2024

Valedictorian, National Chung Hsing University

June 2023

Phi Tau Phi Scholastic Honor Society (top graduate; College of Science representative)

June 2023

Golden Key International Honour Society

Apr 2023

Presidential Award (6 times)

2020 – 2022

Dean's List (2 times)

2020 – 2022

Ching-O Award

Nov 2021

Outstanding Academic Achievement Award

Nov 2021

Professor Kuo Jin-Bin Scholarship

June 2021

Building Futures Foundation Scholarship

Oct 2020

Outstanding Academic Achievement Award

Oct 2020

Certificate of Excellent Performance

July 2020

Skills

Programming: Python, C, Java, MATLAB, JavaScript, C#

Frameworks & Tools: PyTorch, TensorFlow, CUDA, MuJoCo, Unreal Engine, Blender, Django

Languages: Native in Chinese; Fluent in English